[S04] LIME and SHAP: Explaining Machine-Learning Predictions

TOPIC AREA Explainable AI

I. Learning Objectives and Guiding Question

Explain two post-hoc XAI methods and assess the usefulness and stability of their explanations.

Guiding question: Does a highly accurate prediction necessarily come with a reliable explanation?

II. Required Tasks

1. Distinguish explainability from interpretability; local from global explanations; model-specific from model-agnostic methods; and intrinsic from post-hoc interpretability.
2. Explain the LIME workflow: perturbation → locality-based weighting → fitting an interpretable surrogate model. Explain the roles of π_x and $\Omega(g)$.
3. Introduce SHAP through Shapley values and additive feature attribution. Explain the baseline, feature contributions, and the sign of an attribution.
4. Using a public tabular dataset (Iris, Breast Cancer Wisconsin, or a synthetic dataset), train a simple model and select at least three predictions. Explain them with LIME and SHAP, or provide a hand-worked illustration based on a small linear model.
5. Compare the two methods in terms of highest-ranked features, runtime, and sensitivity to random seeds or background data. Include at least one example of an unstable explanation.
6. Discuss why feature importance should not be mistaken for causal effects and propose a procedure for evaluating explanation faithfulness.

III. Equations / Representations to Explain

$$\text{LIME: } \xi(x) = argmin_g\ [L(f, g, \pi_x) + \Omega(g)]$$

$$\text{SHAP: } f(x) \approx \varphi_0 + \sum_j \varphi_j(x)$$

Student need to define every symbol, specify appropriate units/value ranges and modeling assumptions, and provide a complete step-by-step worked example.

IV. Minimum Expected Outputs

A feature-contribution chart for a specific prediction, a LIME-versus-SHAP comparison table, and an example in which explanations differ.

V. Required and Recommended Research Papers

Required: Ribeiro, M. T., Singh, S. & Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. KDD 2016, 1135–1144. https://doi.org/10.1145/2939672.2939778

Reading focus: Study local explanations, perturbation, and interpretable surrogate models.

Required: Lundberg, S. M. & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 2017. https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

Reading focus: Study SHAP values, additive feature attribution, and method limitations.

Recommended: Explainable artificial intelligence: A survey of needs, techniques, applications, and future direction. Neurocomputing 599 (2024), 128111. https://doi.org/10.1016/j.neucom.2024.128111

Reading focus: Review the taxonomy of XAI methods and the approaches used to evaluate explanations.

VI. Format and Submission Requirements

- Team size: three students are recommended. Suggested preparation time: approximately two weeks after the topic is assigned.
- Submit a 6–10-page research-style report in Vietnamese on A4 paper (excluding references and appendices), organized as: Abstract, Introduction, Background/Paper Review, Method/Worked Example, Results and Discussion, Conclusion, and References.

- Prepare 8–12 content slides for a 15-minute seminar presentation followed by approximately 5 minutes of Q&A. Every team member must present and answer at least one question.
- Read all papers marked "Required" in this assignment. Explain the material in your own words and distinguish findings reported in the papers from examples or experiments developed by your team.
- Programming-based experiments are encouraged but not generally mandatory. A worked numerical example, an illustrative model, and an evidence-supported comparison are mandatory.
- Cite all figures, tables, datasets, equations, and AI tools used. Independently verify all claims suggested by AI-generated content.

VII. Seminar Assessment Rubric (100 points)

Assessment criterion	Requirements for full marks	Points
Problem statement and objectives	Establish relevant context, a clear research question, and an appropriate scope.	5
Reading and synthesis of research papers	Accurately explain the original papers' objectives, contributions, methods, and findings, with appropriate source comparison.	15
Explanation of algorithms and equations	Define all variables and assumptions, describe algorithmic steps, and provide at least one correct numerical example or diagram.	15
Evidence, comparison, or experimentation	Include an appropriate baseline or reference, verifiable data or examples, and clearly explained figures and tables.	10
Critical analysis and limitations	Discuss strengths, weaknesses, and conditions of applicability. Avoid overstating conclusions based on a small demonstration.	10
Slide quality and citations	Prepare 8–12 clear, legible slides, cite sources for	10

	figures/equations and relevant papers, and avoid verbatim copying.	
Seminar presentation	Deliver a coherent presentation within 15 minutes, explain examples clearly, and ensure balanced participation.	15
Q&A and critical discussion	Answer questions accurately, justify technical decisions, and ensure that each member answers at least one question.	20

Performance levels for each criterion: Excellent (100%), Good (75%), Satisfactory (50%), and Unsatisfactory (0–25%) of the available points, depending on the quality of evidence submitted.